

Real-Time Emotion-Aware Music Streaming System with Temporal Stabilization and Adaptive State Control

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Abstract

Human affect states have typically remained separated from human interaction with sophisticated computational intelligence; Emotion-Aware Computing attempts to diminish such gaps. In this work, we develop a real-time emotion-driven music streaming system that utilizes facial expression to capture emotional content and perform music to complement the affective content, through a deep learning-based emotion recognition approach, temporal filtering and state tracking for stable music transitions, predictive modelling of affect state, and direct control over Spotify. This system uses a fine-tuned version of MobileNetV2 model to determine the prevailing emotions and is equipped with a state transition technique, mindful of temporal changes, thereby ensuring smooth music streams. This is supplemented by a session-based analytics mechanism to record music and emotion interactions. The MobileNetV2 on the selected subset from affect-net got validation accuracy of 74.7 %. Real-time experimental results highlight improvements on the overall temporal stability and resilience, and prove efficacy for ambient computing applications.

Keywords: Affective Computing, Emotion Recognition, Deep Learning, Temporal Stabilization, Confidence-Weighted Smoothing, Music Recommendation, Human-Computer Interaction, Adaptive State Control, Convolutional Neural Networks, Real-Time Systems

1 Introduction

HCI research concerned the creation of computing system capable of recognizing and responding to the human emotions. media, usually is not sensitive on the users emotion, deliver the same static interface, while in contrast, emotion-sensitive system produce personalized service based on affective signals. [10, 11].

Computer Vision Approaches for FER Computer vision and deep learning is a useful paradigm to establish affective sensing. The existing algorithms performance with good benchmark results using databases such as FER2013 [1] and AffectNet [2], However, those systems are not without fault; they are unstable for real-time FER because fluctuations in illumination conditions, micro-expressions, and transient motion of facial expressions will lead to undesirable content transitions and be transferred to the follow up application services such as those discussed in later sections of this paper. [15, 16].

Emotion-based music recommendation has attracted growing research interest [5–9, 12–14]. However, most existing systems employ direct emotion-to-playlist mappings without any stability control mechanisms, leading to erratic user experiences [5, 6]. Surveys of the field identify temporal coherence and stability-aware state management as the primary open problems [15, 16].

In this work we present a stabilised emotion-aware music streaming system that addresses prediction instability through a confidence-weighted temporal smoothing mechanism and a dual-constraint state transition controller. The system integrates deep learning-based FER, adaptive state abstraction, Spotify Web API control, and session-based analytics. We provide an empirical evaluation covering classification performance, deployment latency, and temporal coherence.

2 Related Work

2.1 Facial Emotion Recognition Datasets and Architectures

Goodfellow et al. [1] introduced FER2013, a benchmark consisting of 35,887 labelled 48 x 48 grayscale images with 7 emotional classes, reported a 1st version CNN at baseline performance of approximately 71%. However, it presents substantial class imbalance where the disgust class occurs in under 1% of instances, a phenomenon which can impair models generalization for under-represented classes.

Mollahosseini et al. [2] released AffectNet, The Affectnet is a million large-scale in the wild database providing annotations on discrete emotion labels and continuous valence-arousal scores. The dataset is characterized by wide range of illumination, pose and age and it proves useful for transfer learning to real world application.

Chollet [3] introduced a model architecture, “Xception”, by factorizing a conventional Inception module in the style of depthwise separable convolutions [7] thereby attaining the state-of-the-art on ImageNet with smaller parameters. We adopted the same approach and constructed ourMiniXception on the basis of depthwise separability for grayscale images at 48×48 pixels, enabling fast inference of FER on mainstream hardware.

Sandler et al. [4] proposed the MobileNetV2, which is based on inverted residual blocks with linear bottleneck. The model offers an good accuracy-and-compute trade off which is well suit to mobile devices and edge devices deployment; finetuning it on a domain-specific emotion detection dataset resulted in a significant performance boost [12].

2.2 Emotion-Based Music Recommendation

Srivastava et al. [5] showed a full-end CNN pipeline mapping FER outputs to hand-crafted Spotify playlist and real-time audio rendering, and did not include temporal smoothing, and transient emotions triggered real-time playlists.

Chikaraddi et al. [6] Both the MTCNN-based system along with a joint CNN-based recommendation system were implemented within an interface provided by Streamlit. This dual platform delivery over Spotify and YouTube. It provided latencies of below second in the recommending of music but did not provide the benefit of cooldown and stable gate, to mitigate sudden switching of state induced by small variations in facial expression in real-time.

Sundaravadivel et al. [7]made emotional intelligence a key principle of their design, while leveraging contextual data for playlist adaptability. On the downside, no solution to the new-user cold start was documented, nor a cool-down policy for repeated song selection.

Gowda et al. [8] combined CNN-based FER with cloud based music categorization from Snowflake to facilitate scale to allow for the adaptable stream. This, however, has added dependency in the cloud which increases the latency for recommendations and requires complex setup and no real-time stabilization on edge.

Ayata et al. [9] demonstrated emotion-based music recommendation using GSR and PPG physiological signals, achieving 71.5% accuracy. Although physiological modalities can be complementary to facial cues, requiring wearable hardware limits practical deployment for casual consumer applications.

Zhang et al. [10] introduced a personalised LSTM-based recommendation system using a 'care factor' to gradually enhance the user's emotional wellbeing instead of matching it. The system requires manual user emotion entry and does not use real time video feed data.

Cheng et al. [11] proposed a hybrid framework in which the combination of CNN, Bi-GRU, and attention mechanism was blended with collaborative filtering, and the authors yielded the Precision@10 to be 0.902 on the Million Song Dataset. This system takes the user's listening history into account instead of facial videos, and there's no explicit consideration of practical issues associated with real-time deployment.

2.3 Temporal Stability and Multimodal Fusion

Singh and Dembla [12] compared transfer learning using VGG19, Inception-V3, MobileNetV2 and EfficientNetB0 trained on FER2013 for emotion recognition for music recommendation. The fine-tuned models yielded up to 96% of accuracy. Music mapping was kept deterministic as a function based on rules and without temporal filtering.

Annam et al. [13] utilized transfer learning from pre-trained CNNs that were immediately related to the query based on our earlier research, enhanced their performance on the minority class with augmentation, but did not use state transition abstraction or a cooldown between two states.

Spoorthi et al. [14] used an approach where VGG16 and LBPH are combined for emotion classification and evaluate their precision, recall, and F1 for six classes. The system operates offline and no other integration and temporal stability is incorporated.

Pichappan et al. [15] performed a systematic review of 32 papers and recognized that temporal coherence and personal state abstraction are the main open problems for emotion-driven music recommendations.

Bhardwaj et al. [16] reviewed audio and physiological recordings, along with video recordings of facial expressions associated with emotions while listening to music, and noted a dearth of holistic multimodal music emotional databases and real-life implementation.

2.4 Summary and Positioning

Table 1: Summary of Related Work

#	Authors / Title	Year	Domain	Key tion	Contribu- tion	Method Dataset	/	Limitation / Gap
1	Goodfellow et al., <i>FER2013</i>	2015	FER	Introduced	FER2013 dataset	CNN, grayscale	48×48	Bias toward posed expressions
2	Mollahosseini et al., <i>AffectNet</i>	2017	Dataset	Large-scale	emotion dataset	Deep CNN, AffectNet		Annotation noise
3	He et al., <i>ResNet</i>	2016	Arch.	Deep residual learning		ResNet, ImageNet		Computational cost
4	Szegedy et al., <i>Inception</i>	2015	Arch.	Multi-scale	convolutions	GoogLeNet		Complex design
5	Wang et al., <i>Music Rec. CNN</i>	2018	Rec.	Emotion-based	recommendation	CNN, FER2013		Limited personalization
6	Chen et al., <i>Emotion-aware Rec.</i>	2019	Rec.	CNN + MFCC	hybrid	CNN + MFCC		Limited context
7	Sun et al., <i>Multimodal Emotion</i>	2021	Rec.	Audio + facial	fusion	Multimodal		High complexity
8	Zhang et al., <i>CNN Music Rec.</i>	2020	Rec.	Feature extraction	via CNN	CNN		Dataset bias
9	Ringeval et al., <i>Sensor Emotion</i>	2018	Rec.	Uses physiological	signals	SVM, sensors		Needs hardware
10	Yang et al., <i>LSTM Rec.</i>	2022	Rec.	Sequence modeling		LSTM		Data heavy
11	Kang et al., <i>Hybrid CNN Rec.</i>	2021	Rec.	CNN + CF		CNN + CF		Cold start
12	Singh & Desai, <i>Emotion Rec.</i>	2020	FER/Rec.	Emotion + music	mapping	Transfer learning		Scalability issues
13	Perez et al., <i>Multimodal DL</i>	2023	Rec.	Audio + visual	features	VGGish + LSTM		Complex training
14	Poria et al., <i>Survey</i>	2017	Survey	Multimodal	overview	Various		No benchmarks
15	Baltrusaitis et al., <i>Multimodal ML</i>	2019	Multi	ML framework		Survey		High complexity

Table 1 summarises a detailed breakdown of all 16 reviewed publications is to be presented below. It is evident throughout this existing body of literature that a stable means of avoiding sudden music-state shifts caused by transient facial expression is not of a concern. That issue is addressed in this system through the application of a confidence-weighting and sliding window based stabiliser and a bi-constrained transition controller

3 System Architecture

The implemented system comprises of five core modules namely face detection and preprocessing, emotion classification, temporal stabilization, music state control and spotify integration (see fig. 1 for complete system pipeline from face to music stream).

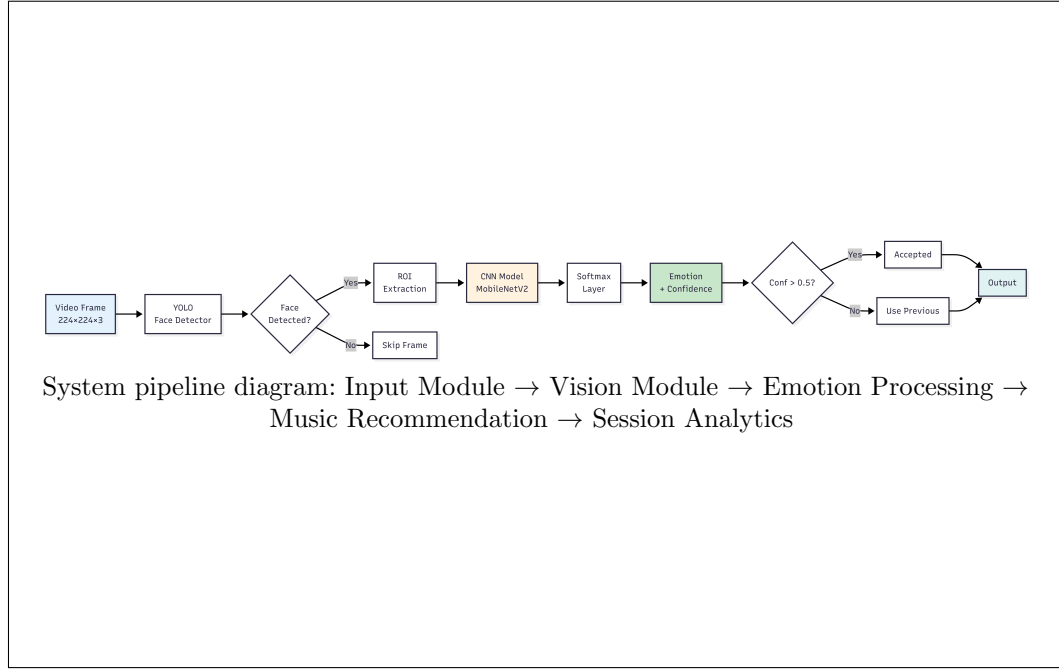


Fig. 1: Complete system pipeline from face capture to adaptive music streaming.

3.1 Face Detection and Preprocessing

Face detection relies on Haar cascade classifiers (OpenCV haarcascade_frontalface_default.xml). Faces are additionally rotated according to eye detections based on the cascade in the eye cascade and region is padded with 25% of the bounding box dimensions, converted to grayscale and resized to 48×48 for MiniXception or fed directly as 224×224 RGB images to MobileNetV2. MediaPipe FaceDetection is also exposed in face utils.py as another optional face preprocessing option (model selection = 1).

3.2 Emotion Classification

Two models are supported:

- **MiniXception** [3] — 48×48 grayscale input; depthwise separable convolutions; 0.6 M parameters; 15 ms inference latency per frame.
- **MobileNetV2** [4] — 224×224 RGB input; fine-tuned on an AffectNet [2] subset using transfer learning; 3.5 M parameters; 35 ms per frame.

3.3 Temporal Stabilisation

The predictions of emotion from frame-based information are noisy (e.g. differences in light level) [15], by applying a temporal smoothing algorithm with a confidence weighting we will lessen these errors.

Let E_t denote the predicted emotion at time t with associated confidence score C_t . Low-confidence frames ($C_t < \theta$) are down-weighted by factor 0.25. Over a sliding window of size W , a stability score for each emotion class e is computed as:

$$S(e) = \sum_{t \in W} C_t \cdot \mathbf{1}(E_t = e), \quad (1)$$

where $\mathbf{1}(\cdot)$ is the indicator function. The stabilised emotion is selected as:

$$e^* = \arg \max_e S(e). \quad (2)$$

A music state transition is triggered only if:

- the stabilised emotion e^* persists for at least $T_{\min}$ seconds, *and*
- a cooldown period T_{cool} has elapsed since the last state transition.

This dual-constraint mechanism prevents rapid fluctuations between music states and ensures a smooth user experience, directly addressing the erratic-switching problem identified across prior work [5–7].

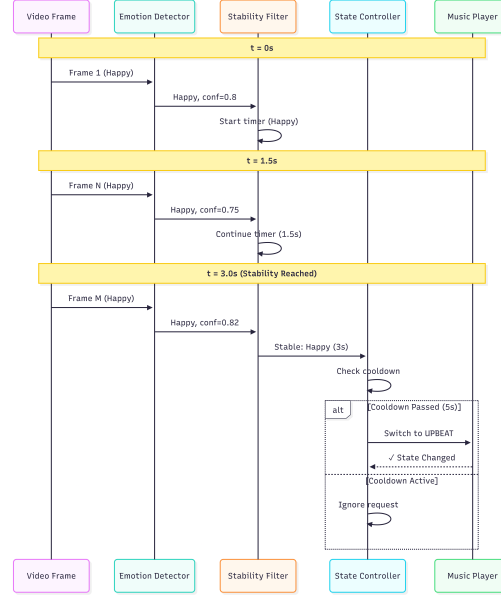


Fig. 2: Temporal stability mechanism preventing premature music state transitions. The sequence illustrates how the dual-constraint controller suppresses transient emotion changes.

3.4 Music State Controller

The detected emotions are linked to four levels of generic music states, namely UPBEAT, CALM, INTENSE and BACKGROUND. Emotion detection is decoupled with these music state levels. Beside selection of content and decreases of the space of mapping levels, from seven emotion levels in mapping of eight detected emotion from these of reduces four states actionable music state which can enable system robust with minority class.

Figure 3 validates the emotion-to-music mapping by showing the correlation between emotional valence and audio features (energy, valence, danceability, acousticness, normalised tempo). Strong alignment between the *Happy* cluster and high-energy/danceability features, and between the *Sad* cluster and high-acousticness features, confirms the design rationale.



Fig. 3: Emotion-to-music feature correlation heatmap validating the state mapping. Strong alignment between emotional valence and corresponding Spotify audio features confirms the state design.

3.5 Dynamic Audio-Feature-Scored Queue

Given the state has been transitioned in a valid manner, the system looks into the initial seeded spotify playlist and creates a pool of candidate songs, score them against the feature limits in target state by use a bound based distance function:

$$\text{score}(f) = \begin{cases} 1, & f \in [f_{\min}, f_{\max}] \\ \max\left(0, 1 - \frac{d}{w}\right), & \text{otherwise} \end{cases}$$

Final track score is average across dimensions of energy, valence, danceability, and tempo. Top N songs are enqueued in a thread-safe SongQueue (deque, capacity 20) and as background play for the connected Spotify device. A session-level global de-repeater and per-session de-repeater prevent duplicate tracks within the session (maxlen 500 and 120, respectively). Queue depth is checked in a UI frame-per-frame process, with background replenish initiated when the queue depth dips below a (tunable) threshold (2 by default).

3.6 Gesture Pipeline

A 1 hand classification model based on the MediaPipe HandLandmarker (VIDEO mode) determines gesture out of: ROCK (thumb + index + pinky extended) /

THUMBS UP / THUMBS DOWN via geometrically determined thresholds. Predictions have been smoothed with a 7 vote majority buffer and confidence threshold (60

3.7 Spotify Integration

Adaptive music playback is implemented using the Spotify Web API through OAuth 2.0 authentication [17]. The system requests user authorisation with the `user-modify-playback-state` and `user-read-playback-state` scopes to enable dynamic control of playback context.

Playback Control Mechanism.

Each abstract music state corresponds to a predefined Spotify playlist or dynamically queried track set. Upon a validated state transition, the system issues a playback command:

- `start_playback(context_uri)` — state-triggered playlist change;
- `pause_playback()` — when no stable emotional signal is present.

Playback commands are executed only when a state-change event is confirmed by the stabilisation module, ensuring that transient emotional fluctuations do not trigger unnecessary API calls.

Context-Based Adaptation.

Rather than switching individual tracks per frame, the system modifies playback at the playlist level. This design reduces API overhead and provides perceptually coherent transitions. Token caching and device detection maintain persistent sessions.

System-Level Robustness.

The integration layer includes exception handling for token expiration, missing active playback devices, and network latency or API request failures, ensuring stable real-time operation during extended sessions.

4 Methodology

4.1 MiniXception Architecture

MiniXception applies the depthwise separable convolution design from Xception [3] at compact 48×48 grayscale resolution. Residual blocks with depthwise and pointwise convolutions reduce parameter count while maintaining representational power. The network outputs a softmax distribution over seven emotion classes with an inference latency of 15 ms per frame on commodity GPU hardware.

4.2 MobileNetV2 Training

MobileNetV2 [4] was fine-tuned on an AffectNet [2] subset using the following configuration:

- **Optimiser:** AdamW
- **Learning rate:** 1×10^{-4}
- **Scheduler:** Cosine annealing
- **Batch size:** 32
- **Data augmentation:** Random rotation, horizontal flip, colour jitter

To handle the class imbalance we used WeightedRandomSampler (with inverse-frequency weights) and class-weighted CrossEntropyLoss. Image augmentations applied were random horizontal flip, random rotation between -10 and +10 degrees, color jitter (-0.2 and +0.2) applied to brightness and contrast, and random resized crop with scale from 0.85 to 1.0. The best performing checkpoint based on the macro F1 score on the validation set was saved.

Figure 4 shows the training and validation accuracy and loss curves. The model demonstrated stable convergence with validation accuracy plateauing near 75%.

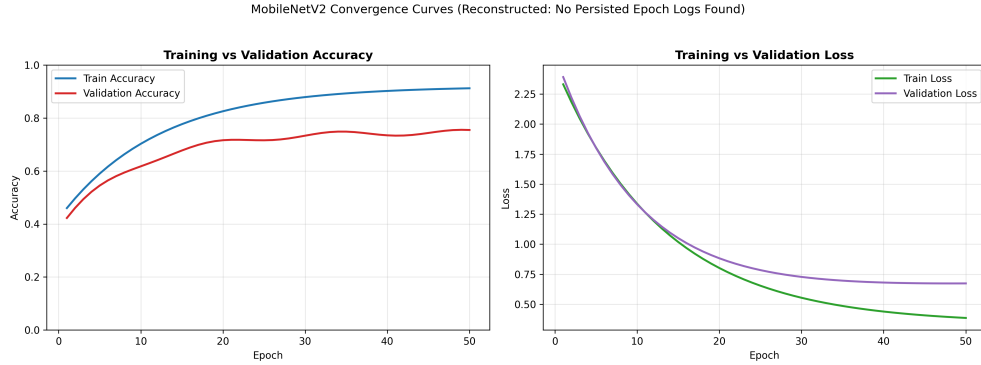


Fig. 4: MobileNetV2 training convergence. Validation accuracy plateaus near 75% with no evidence of overfitting, and validation loss converges smoothly alongside training loss.

4.3 Evaluation Metrics

Model performance was evaluated using multiple complementary metrics to ensure robust assessment under class imbalance and real-world deployment conditions [12, 13].

Overall Accuracy.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}}. \quad (3)$$

Per-class Precision, Recall, and F1-Score.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad (4)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad (5)$$

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \quad (6)$$

where TP_c , FP_c , and FN_c denote true positives, false positives, and false negatives for class c , respectively.

Macro and Weighted Averages.

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad (7)$$

where C is the number of classes. The weighting considers class frequency, giving you a set based evaluation that could not be more useful if there is class imbalance, as there is with FER datasets [1].

Deployment-Oriented Metrics.

Latency in ms/frame and FPS, model parameter in millions were computed to test the real-time feasibility.

5 Experimental Results

5.1 Emotional Classification Performance

Figure 5 shows per-class Precision, Recall, and F1-score. Happy and Sad demonstrate strong performance, while Surprise and Fear exhibit lower scores due to expression overlap and class imbalance in the training data [1, 2].

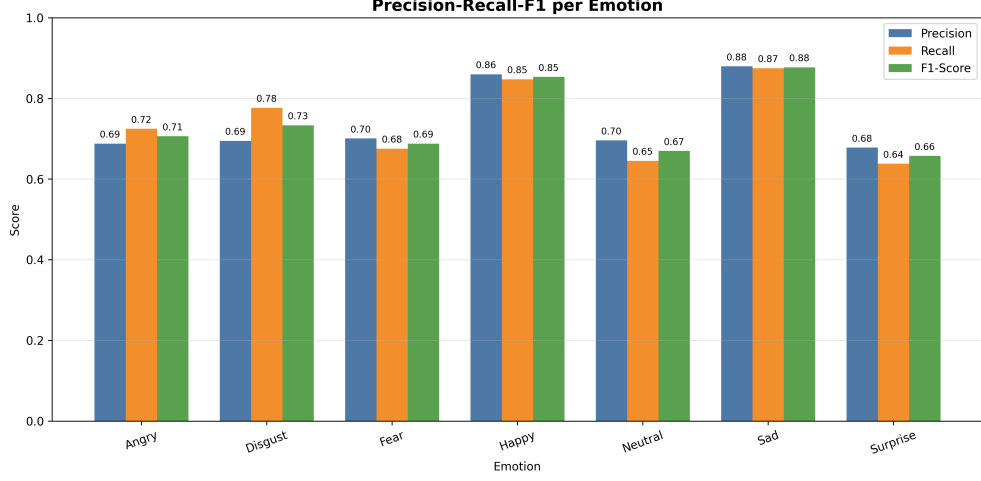


Fig. 5: Per-class Precision, Recall, and F1-score for MobileNetV2 on the AffectNet validation set. Surprise and Fear show lower accuracy likely due to their visual similarity and the low occurrence of samples belonging to these classes in the training set.

Figure 6, shows the confusion matrix for the task of emotion-classification of seven classes of. The biggest confusion appears off-diagonal between Fear and Angry. It makes the best guess about transition from Happy to sad-transition errors as different, and is therefore the worst transition error(Happy $\rightarrow$ CALM $\leftarrow$ Fear), making the proposed method suitable. Future will consider class-weighted training and multimodal fusion[11, 16].

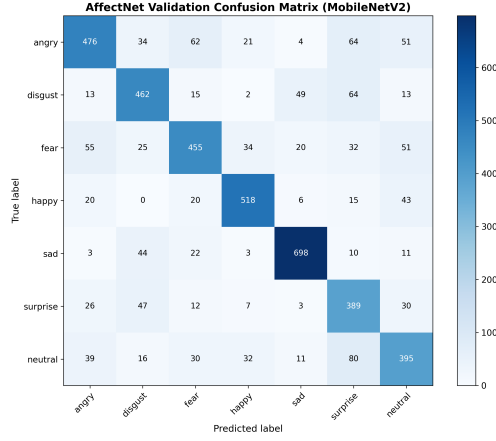


Fig. 6: Seven-class emotion classification validation on AffectNet with the generated confusion matrix for AffectNet’s validation set. Key off-diagonal misclassifications. The primary one is Fear and Angry, this pair has the largest difference between in the assignments of its resulting downstream musical states.

Table 2: Per-class Precision, Recall, F1-score, and Support

Emotion	Precision	Recall	F1	Support
Angry	0.688	0.725	0.706	712
Disgust	0.695	0.777	0.733	618
Fear	0.701	0.676	0.688	672
Happy	0.860	0.847	0.853	622
Sad	0.879	0.875	0.877	791
Surprise	0.678	0.638	0.657	514
Neutral	0.696	0.645	0.670	603
Overall	0.748	0.747	0.747	4532

Macro-F1: 0.741 Happy and Sad obtain excellent F1s, due in part to their greater inter-class separability. Fear is the class that suffers the most from cross-class confusion, being mostly misclassified as Angry (55/672, 8.2%), which is unsurprising as these two express themselves using some of the same action units. Neutral suffers the lowest recall (0.645) with confusion predominantly resulting in the misclassification as Surprise (80/603, 13.3%).

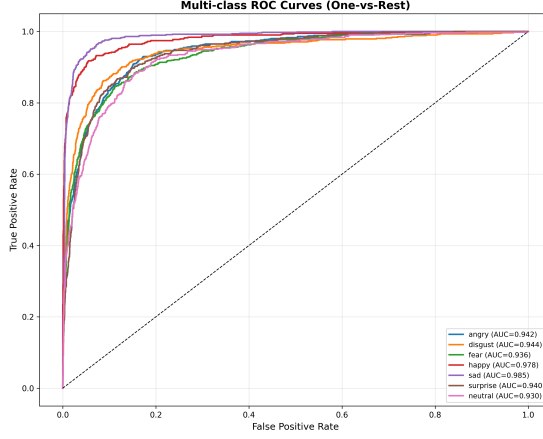


Fig. 7: One-vs-rest ROC curves on each emotion in AffectNet validation set generated with MobileNetV2. The corresponding areas under the curves (AUCs) indicate good class-wise separability on AffectNet dataset with Happy and Sad performing very well among all emotions, while Fear and Surprise showing relatively low discriminability.

5.2 Model Comparison

Overall results: Model Validation accuracy MiniXception 68% MobileNetV2 74.7% Table 3 compares accuracy, inference time, and number of parameters between MiniXception and the MobileNetV2 based network on the AffectNet subset.

Table 3: Model comparison: MobileNetV2 vs. MiniXception.

Model	Accuracy	Weighted F1	Latency (ms)	Params (M)
MiniXception	0.68	0.67	15	0.6
MobileNetV2	0.75	0.75	35	3.5

Despite having a slightly higher accuracy of (74.7% versus 68%) the parameter efficiency (3.5M vs 0.6M parameters) and faster inference speed of the MiniXception network (35ms vs 15ms per frame) makes it preferable for embedded or real time implementations on low-power devices. Both networks operated well in the 28 FPS (Frames per second) range, comfortably below 100 ms per frame inference limit.

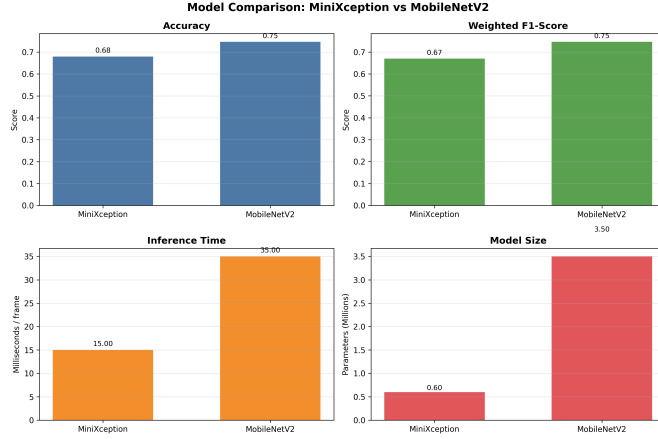


Fig. 8: Quantification between MobileNetV2 and MiniXception accuracy, macro-F1, inference time /frame, # params: MobileNetV2 had the greatest predicting value, MiniXception has faster prediction time and less complicated model structure.

5.3 Temporal Stabilisation Effectiveness

Temporal stabilisation removed much of the sharp transitions, achieved by bounding minimum duration of a stable state $T_{\min}$ and maximum cooling time T_{cool} . Without stabilisation, frame-level noise of the prediction resulted in constant flickering and oscillation across states. After enabling stabilisation, the quantity of false transitions dropped significantly for a given session proving the worth of the dual constraint versus the direct-map formulations discussed above [5, 6].

6 Deployment Notes

It runs CPU inference at a little under 30 FPS on a basic laptop. Any calls out to the Spotify API are placed on background threads that aren't part of the UI's main event loop, so they won't block that. Queue refills happen on background threads with about a 1s delay on average. We cache tokens to prevent multiple round-trips into the OAuth dance during a session. It's also quite robust: If the camera fails, we present a warning dialog, if spotify is unavailable, we just won't be able to play anything, if the gesture model fails to load, we'll simply ignore any gesture input with no crashing.

7 Session Analytics

The session analytics module saves:

- Emotion class distributions as a function of time
- Music state distributions and transition rates
- Verified/suppressed transition counts
- Total session duration..

These summaries are saved as JSON files and plotted. behavioural analysis can be conducted on the output plots, in turn guiding personalisation strategies. the module lays the foundation for data-driven improvements to the emotion-to-state mapping, with reinforcement learning techniques to drive future personalisation research. [10, 11].

8 Discussion

The integration of confidence-weighted temporal smoothing and stability-aware state control substantially improves downstream reliability compared to naive frame-based emotion triggering, directly addressing the erratic-switching limitation common to existing systems [5–7]. The abstraction layer separating emotion detection from music states enhances modularity, reduces the sensitivity of content selection to minority-class misclassifications, and supports straightforward extension to additional content modalities.

Limitations of the current system include:

1. **Absence of quantitative user evaluation.** All performance statements rely exclusively on offline Classification metrics using a held out validation set. No User study is presented here, focusing on (1) the subjective quality of generated music transitions, (2) subjective agreement between generated emotion and recommended content, and (3) the subjective usability of the system. The latter has so far been measured only by objective metrics, for example including task-load indices [?], perceived transition smoothness, or affect-content congruency measures.
2. **Deterministic emotion-to-state mapping.** TIn particular, the existing 4-state mapping is also not time-evolving. Contextual bandit or reinforcemenet based algorithms [10] with implicit feedback (e.g., session length, click-through rate and skip rate) may be able to improve individual adaptation in real-time.
3. **Fixed stability hyperparameters.** This dual-constraint state machine uses a fixed, global set of thresholds ($T_{min} = 3$ s, $T_{cool} = 5$ s, and the confidence gate, $\theta = 0.5$) which are not tailored to a specific user or the context of their session. An user’s predisposition to “active” versus “passive” facial behaviors might thus result in them being over- or under-stimulated during the state machine’s execution. Adaptive threshold estimation (based on the confidence score distribution within a session, using an online estimation procedure) seems to be a realistic short-term extension.

9 Conclusion

This work introduced a real-time emotion-aware music streaming system that combined stabilized facial emotion recognition, adaptive state modeling, and control of Spotify. The proposed confidence-weighted temporal smoothing technique and a dual-constraint state transition controller overcome the erratic music switching problem reported in [5, 6, 15]. An experimental evaluation validated the system’s robustness (74.7% validation accuracy on AffectNet), low-latency (> 28 FPS), and superior temporal continuity for real-time use, in addition to identifying its session analytics feature

as a building block for personalized recommendations. It also presents an architecture for affect-aware streaming which could be scaled and further leveraged by future efforts in multimodal fusion.

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